Roll No.: DL.AI.VYAD24140

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – March 2025

Second Semester

Artificial Intelligence

23AID212 Introduction to IoT

Duration: Two hours

Maximum: 50 Marks

Course Outcomes (COs):

CO1	Familiarize with the fundamental concepts of Internet of Things. .
CO2	Develop skills in programming and hardware platform like Arduino and Raspberry-PI for IOT applications.
CO3	Analyse and integrate the IOT applications to cloud service
CO4	Familiarize with the design and implementation of IOT protocols and connecting devices for IOT application.

Answer all questions

Question No-1 Multiple choice questions

Q. 1A. **How does 5G technology benefit IoT applications?** **[2][CO1][BTL 2]**

Q. 1B. **Name two types of sensors used in IoT and their applications.** **[2][CO1][BTL 2]**

Q. 1C. **Write 3 primary 5G use cases.** **[2][CO1][BTL 2]**

Q. 1D. **What is the main function of the Data Link Layer in the OSI model? [2][CO2][BTL 2]**

Q. 1E. **What is the difference between stop-and-wait and sliding window flow control mechanisms? [2][CO2][BTL 2]**

Question No-2. Explain the ARM processor architecture, its primary features, and its significance in modern computing.
[6][CO1][BTL2]

Question No-3. Write the complete control sequence for the execution of instruction: ADD (R3) R1. You can assume suitable data if required. **[6][CO1][BTL3]**

Question No-4. Explain the different SPI modes in Arduino and describe how they affect data communication. **[6][CO1][BTL3]**

Question No-5. Explain the different types of switching techniques in brief used in data networks and their significance. [6][CO2][BTL2]

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Question No-6. A data network transmits variable-size frames, where the average frame size is 1,200 bytes. The transmission rate is 20 Mbps, and the one-way propagation delay is 6 milliseconds. If the network follows full-duplex communication, compute the total round-trip framing latency, considering both transmission and propagation delays.

[8][CO2][BTL3]

Question No-7. Describe the different types of framing used in packet-switched networks. Explain their significance in brief and how they help in efficient data transmission.
[8][CO2][BTL2].

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	**24**	BTL 2	**12**
		BTL 3	**12**
CO2	**26**	BTL 2	**18**
		BTL 3	**8**

Bloom's Taxonomy Levels (attached for reference)

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create

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